

Joseph Chamoun

Full-Stack Software Engineer & AI Systems Developer

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SUMMARY

Full-stack engineer with internship experience at XpertBot and AI & distributed systems training at Eurisko. Specializes in Laravel, React, and AI integration. Built a production RAG pipeline with FAISS vector search and a Qwen2.5 LLM for a school management system. Experienced in Redis caching, Laravel Sanctum authentication, role-based access control, real-time features (SSE), offline-first mobile architecture, and security testing (OWASP ZAP, Burp Suite). Comfortable working across web, mobile, and AI layers.

EXPERIENCE

Full-Stack Software Engineering Intern — XpertBot Academy

Oct 2024 – Apr 2025 | Remote

- Built **Epic Toy Store**, a full-stack e-commerce platform with a Laravel REST API, Flutter mobile app (users), and a Laravel Blade web store (admin + customers).
- Implemented Redis caching for product listings and user data, API rate limiting, Google OAuth login, and password reset via Mailtrap.
- Built an admin analytics dashboard showing revenue, expenses, best-selling items, and monthly revenue charts with date-range filtering (Chart.js).
- Conducted automated browser testing (Selenium) and vulnerability assessments (OWASP ZAP, Burp Suite) with AWS EC2 hardening.
- Managed Agile workflows via Azure DevOps and Jira.

Software Engineering Trainee — AI & Distributed Systems — Eurisko

Sep 2025 – Oct 2025 | Hybrid | Certificate awarded Jan 2026

- Designed distributed architectures for high-traffic systems: follower graphs, inventory drops, and real-time notifications.
- Built and deployed RAG pipelines using open-source LLMs and FAISS vector search with structured prompt engineering.
- Developed a production-style full-stack MVP with real-time order tracking (SSE) and cloud deployment best practices.

PROJECTS

EduGate — AI-Powered School Management System

[React](#) · [Laravel](#) · [Python](#) · [FAISS](#) · [Hugging Face](#) · [MySQL](#)

AI Repo: github.com/josephchamoun/senior-project-AIassistant

- Built a full RAG chatbot with a **3-layer hybrid intent classifier** (keyword matching → semantic similarity → LLM fallback) and FAISS vector search over school policy documents.
- Ran **Qwen2.5-3B-Instruct** with 4-bit NF4 quantization on a free Colab GPU via FastAPI, reducing VRAM usage from ~12GB to ~2GB.
- Role-based access (student / parent / teacher / admin) enforced at the Laravel API level via Sanctum Bearer tokens — each role sees only their own data.
- School management features: academic years, classes, subjects, schedules, grades, attendance, exams, assignments, and materials — managed by admins and teachers.
- AI assistant designed and built entirely by Joseph Chamoun. Backend and frontend co-developed with Maria Wadih.**

Kfarhawra Brotherhood Platform — Live Production App

[React](#) · [TypeScript](#) · [Laravel](#) · [TanStack Query](#) · [Dexie.js](#) · [InstantDB](#) · [Sanctum](#)

Live: kfarhaoura-brotherhood.vercel.app · github.com/josephchamoun/kfarhawrabrotherhood

- Role-based member management system for a real Lebanese community organization with **20–30 active users**, used daily for events, finances, and records.
- Three sections (Chabiba, Tala2e3, Forsan), each with **8 roles** (President, Amin Ser, Amin Sandou2, Moustachar, etc.) with time-bound assignments and per-role permissions.
- Features: member directory, events with financial tracking, moneyboxes, meetings, elections, shops, and contacts.
- TanStack Query for fast server-state caching; Dexie.js (IndexedDB) for client-side offline persistence on events and members pages.

Epic Toy Store — E-Commerce Platform

[Laravel](#) · [Flutter](#) · [MySQL](#) · [Redis](#) · [Google OAuth](#) · [Sanctum](#) · [Selenium](#) · [OWASP ZAP](#)

[github.com/josephchamoun/internship-web-project](#) · [github.com/josephchamoun/flutter_internship_project](#)

- Full-stack e-commerce platform with a web admin dashboard and a Flutter mobile app for customers.
- Admin dashboard: total orders, revenue, expenses, new customers, best-selling items, and monthly revenue charts — all filterable by date range.
- Redis caching on product listings and user data; API rate limiting; Google OAuth; password reset via email; Laravel Sanctum auth.

Bookstore — Offline-First Android App

[Kotlin](#) · [SQLite](#) · [MySQL](#) · [Firebase Realtime DB](#) · [Android WorkManager](#)

[github.com/josephchamoun/Bookstore_mobile](#) (firebase-migration branch for Firebase version)

- Offline-first Android app with local SQLite storage synced to a MySQL backend — built as part of the Mobile Applications university course.
- **Delta sync** via *updated_at* timestamps — only fetches changed records, minimizing bandwidth on every sync.
- **Background WorkManager** syncs data at scheduled intervals even when the app is closed, ensuring users always see fresh data on launch regardless of last visit.
- Migrated full local storage layer to **Firebase Realtime Database** in a separate branch, gaining hands-on experience comparing local-first vs cloud-first architecture tradeoffs.
- Features: browse and filter books, cart, order placement, favorites, PDF reader for purchased books, and reviews.

SKILLS

Languages	Python, JavaScript, TypeScript, PHP, Java, C#, Dart, Kotlin
Frameworks	Laravel, React, Node.js, Flutter, Kotlin (Android)
AI / ML	FAISS, Hugging Face Transformers, Sentence Transformers, RAG Pipelines, BitsAndBytes
Databases	MySQL, MongoDB, Redis, SQLite, Firebase Realtime DB, InstantDB
Auth & APIs	Laravel Sanctum, Google OAuth, RESTful APIs, SSE
DevOps & Tools	Git, GitHub, AWS EC2, Azure DevOps, JIRA, Postman, Vercel, Render
Testing	Selenium, OWASP ZAP, Burp Suite
Concepts	RBAC, Offline-First Architecture, Delta Sync, Agile, Prompt Engineering
Soft Skills	Leadership, Problem-Solving, Team Collaboration, Public Speaking

EDUCATION

Bachelor of Computer Science — American University of Technology

Graduation: 2026 | Lebanon

CERTIFICATIONS

- **Eurisko:** AI Systems & Full-Stack Engineering Training (Jan 2026) · AI Full Lifecycle for Developers (Oct 2025)
- **XpertBot:** Flutter & Git · Laravel & Git · Data Science · Project Management · Cyber Security · QA
- **Semicolon Academy:** Cybersecurity Workshop (Jan 2026) · Lebanese Universities CTF Competition (Jun 2025)
- **Maxwell Leadership:** Public Speaking & Interview Skills